PoRE: Policy-governed Resolution Engine

Decentralized Source Curation and Deterministic Safety
for Prediction Market Resolution

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Version 1.0 — February 2026

Abstract

Prediction markets have emerged as powerful mechanisms for aggregating information, with platforms like Polymarket surpassing $237M in single-market volume. However, their resolution mechanisms remain critically fragile. Current approaches—most notably UMA Protocol's Optimistic Oracle—delegate final resolution to token-weighted voting, creating structural vulnerabilities where a small number of large token holders can dictate outcomes irrespective of evidence quality. We document three high-profile resolution failures totaling over $247M in affected volume.

We present **PoRE (Policy-governed Resolution Engine)**, a middleware protocol introducing two innovations: (1) a *deterministic policy engine* that mathematically evaluates whether automatic resolution is safe, producing explicit HOLD verdicts with auditable violation reports; and (2) a *Decentralized Source Curation* mechanism that replaces direct outcome voting with token-weighted evidence validation, shifting the manipulation vector from 1-bit outcome corruption to N-dimensional evidence ecosystem subversion. In replay simulations of all three documented failures, PoRE correctly produced HOLD verdicts.

1. Introduction

Prediction markets function as decentralized information aggregation systems where participants express probabilistic beliefs about future events through financial positions [1]. However, a prediction market is only as reliable as its *resolution mechanism*. A corrupted resolution constitutes a direct wealth transfer from honest participants to manipulators.

The current dominant architecture follows a pattern: (1) a proposer asserts the outcome, (2) a challenge window allows disputes, and (3) if disputed, token-weighted voting determines the final outcome. This assumes honest behavior because dishonest voting degrades ecosystem value. We demonstrate this assumption fails when the value at stake exceeds expected token value loss, and when low voter participation amplifies whale influence.

PoRE introduces a *deterministic, non-manipulable safety layer*. Rather than asking "What is the answer?", PoRE asks "**Is it safe to auto-resolve?**" Furthermore, we propose **Decentralized Source Curation**—token holders vote on which evidence sources are valid, and an AI policy engine evaluates the curated evidence deterministically.

2. Problem Statement

2.1 The Resolution Trilemma

Prediction market resolution faces a trilemma among *decentralization*, *accuracy*, and *speed* (Figure 1). Existing systems optimize for decentralization + speed (accepting accuracy risk). PoRE shifts the curve: auto-resolving clear cases (speed), escalating ambiguous ones (accuracy), without central authority (decentralization).

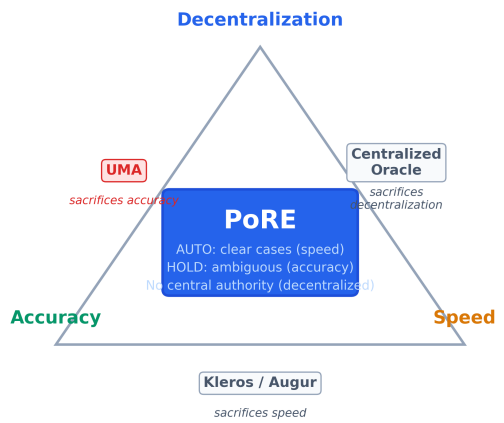

Figure 1: Resolution trilemma. PoRE uses conditional routing (AUTO/HOLD) to approach all three vertices.

2.2 Structural Vulnerabilities

The UMA Optimistic Oracle [3] operates through proposal, challenge, and DVM voting. We identify three vulnerabilities:

V1: Plutocratic determination. With ∼120M UMA supply and 10–20% participation, a holder of 5M tokens controls 20–40% of effective voting power.

V2: Economic misalignment. The security condition requires:

$$C_{\text{manip}} > P_{\text{manip}} \tag{1}$$

For the Ukraine case ($7M market):

$$\begin{aligned} &C \approx 5\% \times \$10\text{M} = \$500\text{K} \\ &P \approx \$1\text{M to } \$5\text{M} \\ &P > C \implies \text{manipulation is rational} \end{aligned} \tag{2}$$

V3: Semantic ambiguity. Natural language criteria create interpretation gaps ("suit" vs. "blazer").

2.3 Documented Failures

Table 1 summarizes three cases totaling $247M+ in affected volume.

Table 1: Documented resolution failures.

Case	Vol.	Exploit
Ukraine mineral deal	$7M	Whale 25%
Zelenskyy suit	$237M	Ambiguity
Fort Knox gold	$3.5M	UI disabled

3. Related Work

Chainlink [2] provides decentralized oracle networks optimized for objective numerical data, not subjective events. **UMA** [3] is efficient for uncontroversial resolutions but degenerates to plutocratic voting. **Kleros** [4] uses random juries but introduces latency. **Augur v2** [5] uses escalating bonds, exploitable by well-funded attackers.

PoRE is a **middleware safety layer**:

$$\text{Sources} \rightarrow \boxed{\text{PoRE}} \rightarrow \begin{cases} \text{AUTO} & \text{resolve} \\ \text{HOLD} & \text{escalate} \end{cases}$$

4. System Architecture

4.1 Five-Layer Design

PoRE consists of five layers (Figure 2):

1. **Evidence Collection** (pluggable): on-chain feeds, APIs.
2. **Source Curation** (decentralized): token-weighted validation.
3. **AI Deliberation** (probabilistic): proposer/challenger.
4. **Policy Engine** (deterministic): 8 hard safety rules.
5. **On-chain Recording** (immutable): verdict hashes.

The design principle is *separation of trust domains*: humans curate (L1–2), AI interprets (L3), code evaluates (L4), blockchain records (L5). No single layer has complete authority.

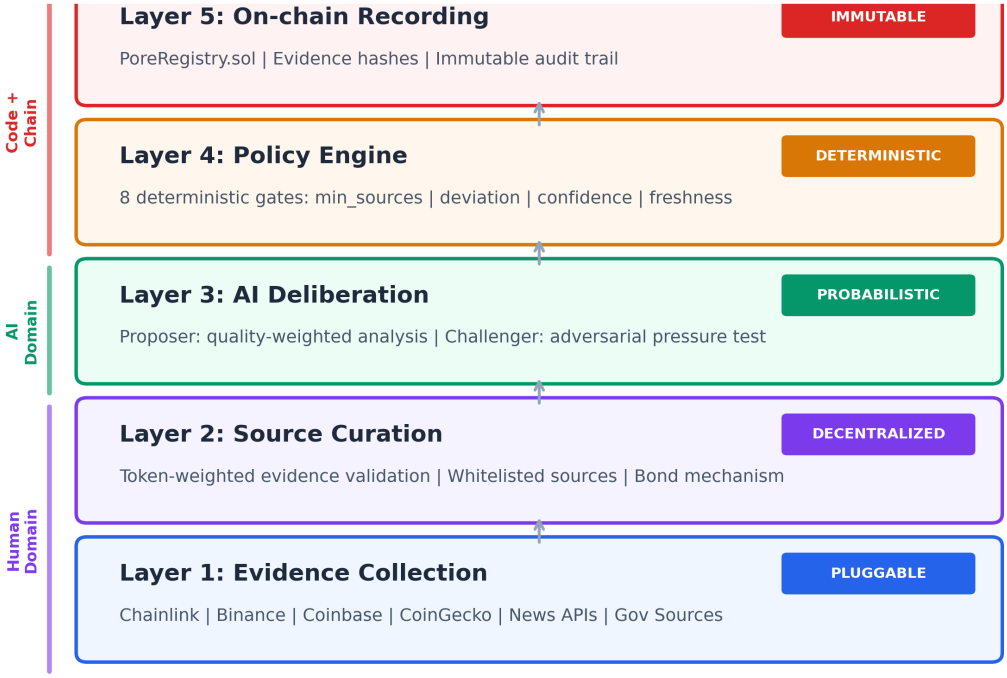

Figure 2: PoRE five-layer architecture. Trust domains: humans curate evidence (L1–2), AI interprets (L3), code evaluates safety (L4), blockchain records (L5).

4.2 State Machine

Each resolution progresses through defined states (Figure 3):

$$\text{PENDING} \rightarrow \text{PROPOSED} \rightarrow \text{CHALLENGED} \rightarrow \{\text{ACC.}/\text{REJ.}\} \xrightarrow{\pi} \{\text{AUTO}/\text{HOLD}\}$$

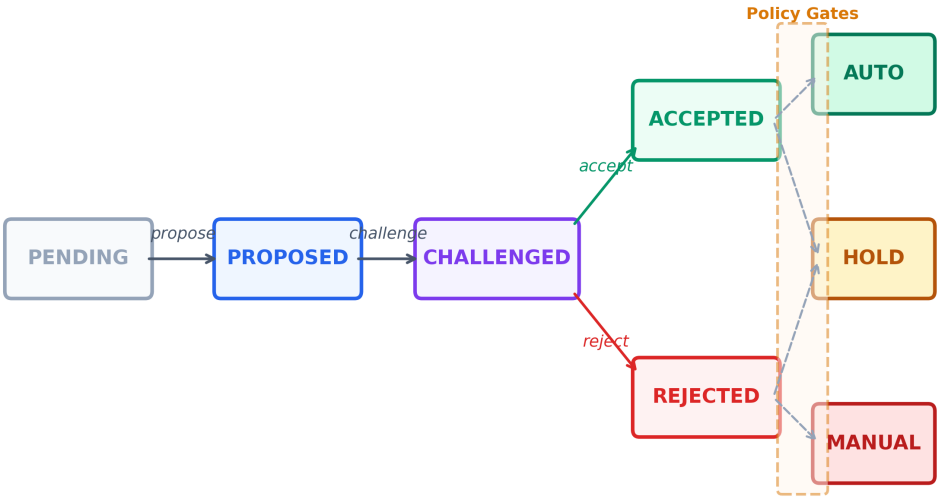

Figure 3: State machine. Policy gates (dashed) are the final safety barrier. All transitions are deterministic and logged.

4.3 Evidence Schema

Each evidence item e_i is a tuple:

$$e_i = (s_i,\ \tau_i \in \mathcal{T},\ q_i \in [0,1],\ v_i \in [0,1],\ h_i) \quad (3)$$

where s_i is the source, τ_i is the type, q_i is quality, v_i is the YES-signal, and h_i is the SHA-256 hash.

5. Deterministic Policy Engine

The policy engine is the *trust anchor*—deterministic code, not AI. Given identical inputs, it produces identical outputs.

5.1 Policy Parameters

A policy π defines:

$$\pi = (n_{\min},\ t_{\min},\ \delta_{\max},\ \gamma_{\min},\ \alpha_{\max}) \quad (4)$$

where $n_{\min}$ = min sources, $t_{\min}$ = min source types, $\delta_{\max}$ = max deviation, $\gamma_{\min}$ = confidence threshold, $\alpha_{\max}$ = max evidence age (hours).

5.2 Evaluation Algorithm

Algorithm 1 Policy Evaluation

Require: Evidence $E = \{e_1, \ldots, e_n\}$, policy π
Ensure: Verdict, violation set V

$V \leftarrow \emptyset$
if $|E| < n_{\min}$ **then**
  $V \leftarrow V \cup \{\texttt{MIN_SRC}\}$
end if
if $|\{\tau_i\}| < t_{\min}$ **then**
  $V \leftarrow V \cup \{\texttt{TYPE_DIV}\}$
end if
$\bar{v} \leftarrow \sum_i q_i v_i \ / \ \sum_i q_i$ ▷ Weighted mean
$\sigma \leftarrow \sqrt{\sum_i q_i (v_i - \bar{v})^2 \ / \ \sum_i q_i}$
if $\sigma > \delta_{\max}$ **then**
  $V \leftarrow V \cup \{\texttt{DEVIATION}\}$
end if
$\gamma \leftarrow f(\sigma, E)$ ▷ Eq. 5
if $\gamma < \gamma_{\min}$ **then**
  $V \leftarrow V \cup \{\texttt{LOW_CONF}\}$
end if
if $V = \emptyset$ **then return** AUTO
else return HOLD, V
end if

5.3 Confidence Score

$$\gamma = w_1(1-\sigma) + w_2 \frac{|\mathcal{T}_E|}{|\mathcal{T}|} + w_3 \left(1 - \frac{\max_i \text{age}(e_i)}{\alpha_{\max}}\right) \quad (5)$$

with default weights $(w_1, w_2, w_3) = (0.5, 0.3, 0.2)$.

5.4 Why Not AI?

LLMs are stochastic (different outputs for identical inputs), unauditable ("the AI felt insufficient" is not verifiable), and adversarially vulnerable (prompt injection). The inequality $\sigma > 0.30$

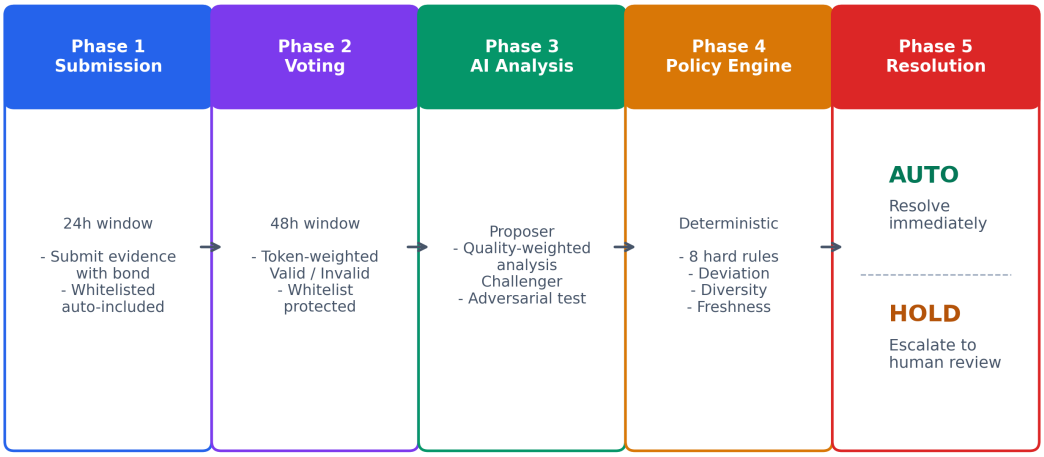

Figure 4: Decentralized Source Curation protocol. Token-weighted validation (Phase 2) feeds into deterministic policy evaluation (Phase 4).

is verifiable by anyone, always.

6. Decentralized Source Curation

This section presents the primary novel contribution.

6.1 Core Idea

Current systems: token holders vote on outcomes (1-bit). We propose voting on *evidence validity*:

$$\text{UMA:} \quad \text{voters} \xrightarrow{\text{vote}} \{\text{YES}, \text{NO}\}$$

$$\text{PoRE:} \quad \text{voters} \xrightarrow{\text{curate}} E' \xrightarrow{\text{engine}} \text{verdict}$$

6.2 Protocol Phases

Five phases (Figure 4):

1. **Submission** (24h): anyone submits with bond b.
2. **Voting** (48h): token-weighted VALID/INVALID. Whitelisted types $\mathcal{W}$ cannot be excluded.
3. **AI Analysis**: proposer + challenger.
4. **Policy Engine**: Algorithm 1.
5. **Resolution**: AUTO or HOLD.

6.3 Quality Scoring

Base quality $q_0(\tau)$ (Table 2) is adjusted by vote ratio r_i:

$$q_i = q_0(\tau_i) \cdot (0.8 + 0.4 \cdot r_i) \tag{6}$$

Even with unanimous support ($r_i = 1$):

$$q_{\texttt{social}} = 0.20 \times 1.20 = 0.24 \ll 0.95 = q_{\texttt{oracle}} \tag{7}$$

Table 2: Base quality by source type.

Source Type	q_0
`onchain_oracle`	0.95
`official_govt`	0.95
`news_agency`	0.85
`cex_spot`	0.75
`aggregator`	0.60
`image_analysis`	0.50
`social_media`	0.20

6.4 Whitelist Mechanism

Types $\mathcal{W} = \{\texttt{official_govt}, \texttt{onchain_oracle}\}$ cannot be excluded by vote. If the State Dept publishes "No deal," this source persists regardless of token holder preferences.

6.5 Anti-Manipulation

Flooding: bond per submission, deduplication, type diversity rule, quality ceiling. **Removal**: whitelist, exclusion metadata, min source count. **Sybil**: token-weighted voting inherently resists sybil. **Collusion**: policy engine independently checks all constraints.

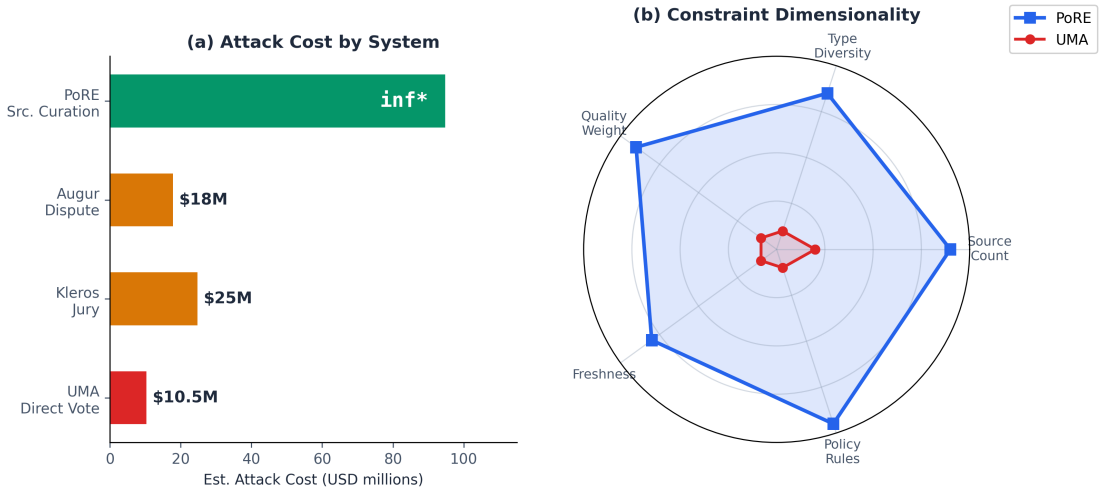

Figure 5: (a) Attack cost comparison. PoRE with whitelisted sources is structurally infeasible. (b) Constraint dimensionality radar.

7. Game-Theoretic Analysis

7.1 Attack Cost

Under UMA (Figure 5a):

$$C_A = \max\left(b,\ \frac{n_{\text{maj}} \cdot p_{\text{tok}}}{\text{participation}}\right) \approx \$10.5\text{M} \tag{8}$$

Under PoRE, the attacker must satisfy constraints (9)–(13) simultaneously:

$$|E'| \geq n_{\min} \tag{9}$$
$$|\mathcal{T}_{E'}| \geq t_{\min} \tag{10}$$
$$\sigma(E') \leq \delta_{\max} \tag{11}$$
$$\gamma(E') \geq \gamma_{\min} \tag{12}$$
$$\forall\, e_w \in \mathcal{W}:\ e_w \in E' \tag{13}$$

Constraint (13) is decisive: whitelisted sources reflecting ground truth cannot be removed. When they contradict the desired outcome, satisfying (11)–(12) is infeasible:

$$C_B \to \infty \tag{14}$$

7.2 Dimensionality

Table 3: Manipulation dimensionality.

System	Target	Dim.
UMA	YES/NO	1 bit
Kleros	Juror verdicts	N bits
Augur	Dispute bonds	Sequential
PoRE	**Evidence set**	$\mathbf{N} \times \lvert\mathcal{T}\rvert \times \mathbb{R}^3$

7.3 Nash Equilibrium

The **honest equilibrium** is dominant: voters validate truthfully, clear cases resolve via AUTO, ambiguous cases escalate. The manipulation equilibrium is unstable: high cost, uncertain success, and full traceability (excluded sources logged on-chain).

8. Case Studies

8.1 Ukraine Mineral Deal ($7M)

A whale (∼5M UMA, ∼25% voting) forced YES with no official confirmation. PoRE replay at T+1h (Figure 6):

Quality-weighted average:

$$\begin{aligned}\bar{v} &= \frac{\sum_i q_i v_i}{\sum_i q_i} \\ &= \frac{0.095+0.143+0.180+0.170+0.158}{3.85} \\ &= 0.194 \quad \text{(NO)}\end{aligned} \tag{15}$$

Deviation $\sigma = 0.43 > 0.30$. Five violations. **Verdict: hold.**

Under source curation, State Dept and Ukraine MFA are whitelisted ($q = 0.95$), dominating the weighted average. Even flooding with social media ($q_{\max} = 0.24$) cannot shift $\bar{v}$.

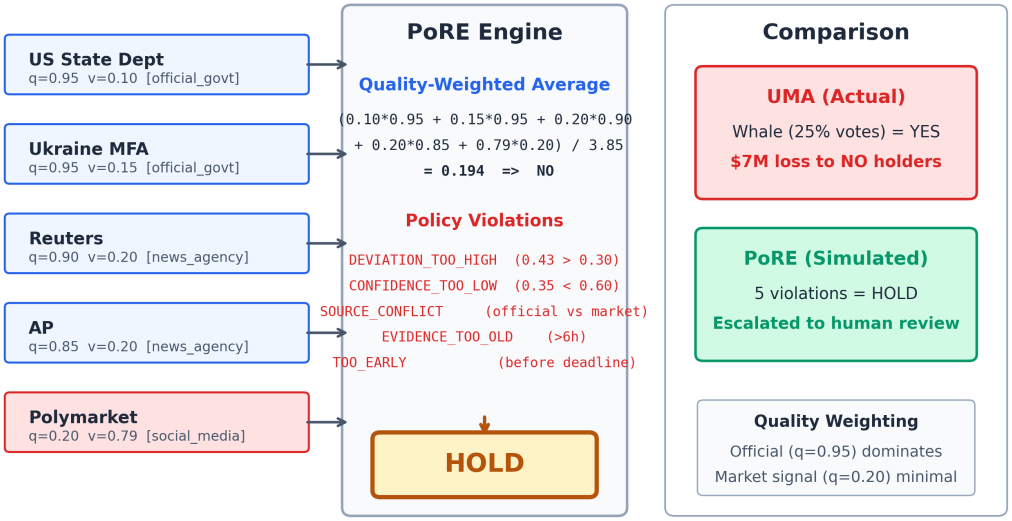

Figure 6: Ukraine mineral deal case. Official sources (q=0.95) dominate the weighted average, producing five violations and HOLD. UMA resolved YES via whale voting ($7M loss).

8.2 Zelenskyy Suit ($237M)

Photo evidence from Reuters/AP/Getty, but two whales resolved NO. PoRE: $\bar{v} = 0.718$ (YES-leaning), $\sigma = 0.38 > 0.30$. Source type conflict. **Verdict: hold.** Correct—genuinely ambiguous.

9. Implementation

The prototype is JavaScript (Node.js), zero dependencies beyond `ethers`:

Table 4: Core modules.

Module	Function
`policy/evaluator`	Deterministic engine
`ai/proposer`	Quality-weighted analysis
`ai/challenger`	Adversarial challenge
`evidence/chainlink`	Mainnet price feed
`evidence/liveCollect`	4-source real-time
`workflow/onchain`	On-chain recording
`PoreRegistry.sol`	Solidity registry

Live evaluation collects from Chainlink BTC/USD (Ethereum mainnet), Binance, Coinbase, and CoinGecko. All 9 tests pass; self-score 100/High.

10. Discussion

Limitations. Evidence collection for subjective events uses pre-curated data; production needs automated pipelines. Base quality scores are expert-assigned; empirical calibration is future work. Policy parameter governance itself requires a mechanism.

vs. Pure AI. A well-prompted LLM approximates PoRE *most of the time*. For $237M, "most" is insufficient. LLMs are stochastic, unauditable, and adversarially vulnerable. PoRE's policy layer is code: same input, same output, always.

11. Future Work

1. Smart contract for source curation voting.
2. Empirical quality calibration from historical data.
3. Cross-platform federation (Polymarket, Kalshi, Augur).
4. Formal verification of policy engine properties.
5. Proof-of-personhood hybrid voting (World ID).

12. Conclusion

Prediction markets need a system that knows *when it doesn't know*. PoRE introduces a deterministic policy engine that cannot be manipulated, and a source curation protocol shifting manipulation from 1-bit corruption to N-dimensional constraint satisfaction. In replays of $247M+ in failures, PoRE produced HOLD in every case.

The oracle problem is solved not by more powerful AI or larger token stakes, but by architectural separation: AI interprets, code enforces, humans arbitrate, blockchain records.

References

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